

Anusha Mahesh

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SUMMARY

Computer Science graduate student at RIT with experience building AI-powered, cloud-native software systems—from federated learning models in PyTorch to serverless NLP pipelines on AWS. Proficient in Python, distributed system design, and integrating AI services into end-to-end software pipelines. Seeking a Fall 2026 internship to contribute to innovative projects in AI and cloud infrastructure.

EDUCATION

Rochester Institute of Technology (RIT), Rochester, NY

Jan 2025 - May 2027

Master of Science in Computer Science (GPA: 3.76 / 4.00)

Relevant Coursework: Foundations of Artificial Intelligence, Machine Learning, Computer Vision, Algorithms & Complexity, Engineering Cloud Software Systems, Visual Analytics

Technical Interests: Machine learning systems, cloud-native software architecture, AI infrastructure, distributed systems

SKILLS

Programming Languages: Python, Java, C++, SQL

AI & Machine Learning: Federated Learning, Computer Vision, Classification, Model Evaluation, PyTorch, scikit-learn, OpenCV, NumPy, Pandas, NLP, Amazon Comprehend

Software Engineering & DevOps: Git, GitHub Actions, Docker, Terraform, CI/CD, REST APIs, Serverless Architecture

Cloud (AWS): EC2, S3, Lambda, DynamoDB, API Gateway, Cognito, CloudFront, CloudWatch, SNS, IAM

Data Engineering: Data Pipelines, ETL, Data Processing, Data Modeling, Data Visualization

Visualization & Analytics: Tableau, Power BI, Plotly, Matplotlib, Seaborn, Gephi

PROJECTS

TrendPulse – PR Sentiment Analysis & Crisis Detection

Jan 2026

Graduate Project | Rochester Institute of Technology

- Built an AI-powered serverless AWS platform that applies Amazon Comprehend NLP to Google News feeds to detect real-time PR crises
- Architected an event-driven, serverless backend with AWS Lambda, S3, DynamoDB, and API Gateway, decoupling RSS ingestion, NLP processing, and persistence into independently scalable components.
- Developed a React dashboard secured with AWS Cognito, displaying real-time sentiment scores, risk categories, and article-level details for monitored brands
- Automated infrastructure and CI/CD processes with Terraform and GitHub Actions, and set up CloudWatch metrics and SNS alerts for production monitoring and automated crisis notifications

Personalized Federated Learning (FedPer) for Non-IID Data

Dec 2025

Graduate Project | Rochester Institute of Technology

- Implemented a Personalized Federated Learning (FedPer) framework in PyTorch, training ResNet-34 on CIFAR-10 across 8 distributed clients with shared base layers and personalized heads to handle non-IID data
- Developed custom aggregation methods that reduced client-server communication by about 15%, improving scalability
- Achieved around 86% accuracy under non-IID data, showing the approach is robust in heterogeneous environments

EXPERIENCE

Machine Learning Intern

Aug 2023 - Oct 2023

Prinston Smart Engineers | Bengaluru, India

- Developed Python-based classification models to predict cancer risk from clinical patient data, including building preprocessing pipelines to handle missing values, outliers, and feature scaling
- Performed feature engineering and exploratory analysis on 1,000+ patient records using Pandas and NumPy to identify key predictors
- Trained and evaluated Random Forest and SVM classifiers, selecting the best model based on accuracy, precision, recall, and F1-score